Dataset and question: I will examine daily maximum, minimum, and mean air temperature at the St. John's International Airport, Newfoundland and Labrador, from the original data provided by the Environment and Climate Change Canada (ECCC). Station ID 50089 (ST. JOHN'S INTL A; climate ID 8403505) provides 365 daily records for 2025. I will explore differences in temperature during the different seasons of the year and in daily temperature range and how the number of days at the extremes of temperature would be affected by the definition. There are cover checks, I plan to get 2015-2024 observations from the same station as a reference.

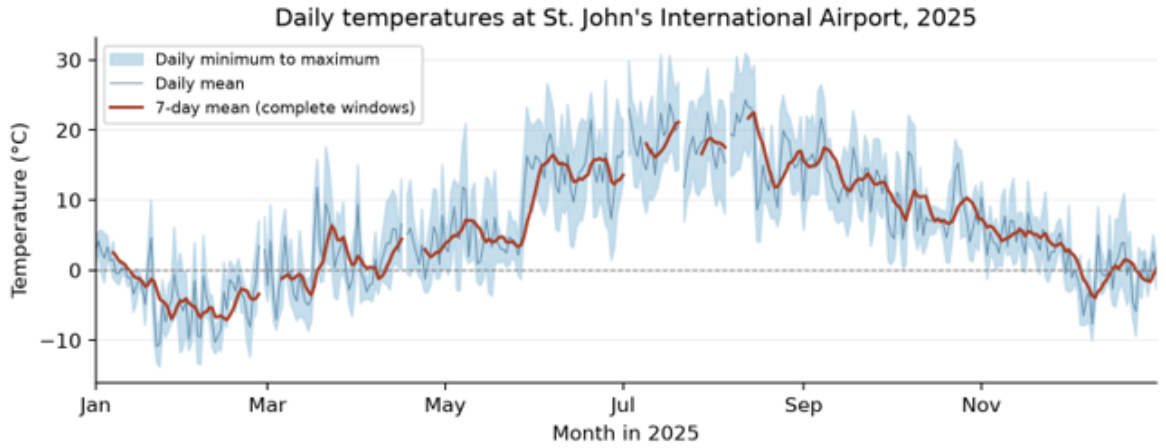


Figure 1. Observed daily temperatures in 2025. The shaded band spans the daily minimum and maximum; the blue line is the reported daily mean and the red line is a trailing seven-day mean requiring complete windows. Missing temperature records (Max Temp: 5, Min Temp: 5, Mean Temp: 5) are not replaced with zero. This first plot describes seasonal variation, not a long-term climate trend.

function and analysis: The main function I am going to derive from my primary data will be the daily temperature range, DTR = max temperature - min temperature. I will test this function with typical numbers, missing data, and invalid pairs of maximum and minimum numbers. Temperature and DTR will be summarized and plotted monthly. I'll audit the reference data, then identify unusually warm days as those that record a maximum temperature above the reference percentile for the top 95th percentile of the calendar month, and unusually cold days as those with a minimum temperature below the reference percentile for the bottom 5th percentile of the calendar month. I will report the number and fraction of valid observations and repeat the analysis at the 90th/10th and 97.5th/2.5th percentiles to test sensitivity.
Trends and deliverables: I will examine the seasonal and, where possible, annual mean temperature trends over the last 11 years (2015-2025) as well as the seasonality. Climate conclusions cannot be drawn from the short record; the ten-year reference is not a climate normal. Notes about missingness and quality flags will be made. The report will be 8-10 pages long, and the presentation will be 5 minutes long, while using Python scripts to create the 6-8 figures, testing the analysis functions, and creating incremental Git commits. The scope is one station and analysis of temperature.

Source: Environment and Climate Change Canada, Historical Climate Data, daily data for station 50089, 2025. Accessed October 20, 2026. Historical Data - Climate - Environment and Climate Change Canada . The exact download URL and raw data are retained with the project